

Parameter Estimation and Optimal Control in Epidemic Modelling

Amartya Amritanshu

Roll No.: BSDCC2405

Centre: Indian Statistical Institute, Kolkata

Email: bsdcc2405@mail.isical.ac.in

April 2026

Contents

1	Abstract	2
2	Introduction	2
3	Objective	2
4	Literature Review	3
5	Data Collection	3
6	Mathematical Modelling	4
7	Methodology	6
7.1	Environment Setup	6
7.2	Data Collection and Ingestion	6
7.3	Data Preprocessing	7
7.4	Exploratory Data Analysis	7
7.5	SIR Model Development	8
7.6	Model Evaluation	9
8	Results	9
8.1	Global Cumulative Trends	9
8.2	Daily New Cases and Deaths	9
8.3	Country-Level Distribution	10
8.4	Cross-Country Correlation	10
8.5	SIR Model Fit	11
8.6	SIR Compartment Trajectories	12
8.7	Residual Analysis	12
9	Discussion	13
10	Conclusion	14

1 Abstract

This project investigates the spread of infectious diseases using the SIR (Susceptible–Infected–Recovered) epidemic model, with a focus on COVID-19. Real-world data sourced from the Johns Hopkins COVID-19 database is used to perform exploratory data analysis and to estimate key model parameters, namely the transmission rate β and the recovery rate γ . The estimated basic reproduction number $R_0 = \beta/\gamma$ characterizes the contagiousness of the disease. Residual analysis is conducted to evaluate model fit. The project further discusses the effect of intervention strategies such as vaccination and social distancing through the introduction of a control variable. Results show that early intervention significantly reduces the infection peak, and that the SIR model provides a tractable framework for public health decision-making despite its simplifying assumptions.

2 Introduction

The COVID-19 pandemic emerged in late 2019 and within months had spread to every corner of the world. By the end of 2022, it had produced over 670 million confirmed cases and nearly 7 million recorded deaths, putting healthcare systems under severe pressure and forcing governments into difficult decisions about lockdowns, vaccines, and other interventions. Against that backdrop, being able to model how a disease spreads — and how much difference an intervention actually makes — became genuinely important.

Mathematical models have long been a useful tool in epidemiology. The basic idea is to divide a population into compartments (say, susceptible, infected, and recovered), write down equations for how people move between them, and use the resulting system to understand the shape of an outbreak. The simplest and most widely used version is the SIR model, introduced by Kermack and McKendrick [1] back in 1927, which still forms the foundation of most epidemic modelling done today.

This project applies the SIR model to global COVID-19 data from the Johns Hopkins University CSSE repository [2]. There are two main things we set out to do. First, estimate the transmission rate β and recovery rate γ from the observed case data — since these can't be solved for analytically, this is done numerically using `scipy.integrate.odeint` and a least-squares optimiser. Second, look at what happens theoretically when an intervention like social distancing or vaccination is introduced, modelled through a control variable $u(t)$ that reduces the effective transmission rate to $\beta(1 - u(t))$.

Along the way the project also covers a full exploratory analysis of the global and country-level trends, a cross-country correlation study, and a residual analysis of the model fit — all of which give a clearer picture of what the SIR model captures well and where it falls short.

3 Objective

The main objectives of this project are:

- To study the spread of infectious diseases using an SIR epidemic model
- To understand the roles of susceptible, infected, and recovered populations
- To estimate important model parameters such as the transmission rate β and the recovery rate γ
- To compare the model predictions with observed epidemic data
- To analyze how changes in parameters affect the spread of the disease
- To study the effect of intervention strategies such as vaccination or social distancing
- To compare the epidemic with and without control measures
- To understand how mathematical modelling can help in public health decision-making

4 Literature Review

Compartmental models. The foundations of mathematical epidemiology were laid by Kermack and McKendrick [1], whose seminal 1927 paper introduced the SIR framework and derived the epidemic threshold condition: an outbreak can only grow if the initial reproduction number $R_0 > 1$. The model remains the standard starting point for epidemic analysis nearly a century later due to its analytical tractability and interpretability.

Extensions of the basic SIR model have been developed to accommodate more realistic biological features. The SEIR model adds an *Exposed* compartment for individuals who are infected but not yet infectious, making it more appropriate for diseases with a latent period such as COVID-19 (median incubation ≈ 5 days [3]). Models with vital dynamics (births and deaths), age structure, spatial heterogeneity, and network-based contact patterns have also been studied extensively [4].

Parameter estimation for COVID-19. A large body of work emerged rapidly after the onset of the pandemic to estimate R_0 and the time-varying effective reproduction number R_t for SARS-CoV-2. Early estimates for the original Wuhan strain placed R_0 between 2.2 and 3.6 [5]. Subsequent variants were found to be more transmissible: Delta ($R_0 \approx 5$ –6) and Omicron ($R_0 \approx 8$ –15) posed substantially greater challenges to control [6]. Methods used for estimation include maximum likelihood, Bayesian Markov chain Monte Carlo (MCMC), and particle filtering on state-space models.

Numerical methods. Because the SIR system has no closed-form solution in the time domain (see Section 6), numerical integration is essential. Dormand and Prince [7] introduced the RK45 adaptive step-size scheme widely used today; SciPy’s `odeint` uses the LSODA algorithm [8], which adaptively switches between Adams (non-stiff) and BDF (stiff) methods, making it well-suited to epidemic ODEs whose stiffness changes as the epidemic evolves.

Optimal control. Optimal control theory applied to epidemic models dates to the work of Sethi and Staats [9] and has expanded significantly since. The objective is typically to minimise a cost functional balancing total infections against the economic cost of interventions. Pontryagin’s maximum principle yields necessary conditions for the optimal control $u^*(t)$. Applied to COVID-19, several studies have shown that *early, moderate* sustained intervention outperforms both no intervention and delayed severe lockdowns in terms of total mortality and economic cost [10].

5 Data Collection

The dataset used in this project is the Johns Hopkins University Center for Systems Science and Engineering (JHU CSSE) COVID-19 repository [2], one of the most comprehensive and widely cited open-access sources of pandemic data. The repository was maintained from January 2020 until March 2023, when data collection was officially discontinued.

The specific files used are the “CONVENIENT” format CSVs:

- `CONVENIENT_global_confirmed_cases.csv` — daily new confirmed cases for every country and sub-national region
- `CONVENIENT_global_deaths.csv` — daily new deaths for every country and sub-national region

In this format, each column corresponds to a country or region and each row to a calendar date, with cell values representing the number of *new* events reported on that day (not cumulative totals). Sub-national regions are denoted by suffixes (e.g. `Australia.1` for New South Wales), which were aggregated into national totals during preprocessing.

The dataset spans from 22 January 2020 to 9 March 2023, covering 1143 days and 201 countries and territories. After preprocessing and aggregation, the global time series yielded the following headline statistics:

- Total confirmed cases: approximately 670 million
- Total confirmed deaths: approximately 6.9 million
- Case Fatality Rate (CFR): approximately 1.03%

- Peak single-day global new cases: approximately 3.5 million (January 2022, Omicron wave)

It is important to note that confirmed case counts are a significant underestimate of true infections, as they depend heavily on testing capacity, which varied enormously across countries and over time. Deaths data is generally considered more reliable, though also subject to reporting delays and definitional differences between countries. These limitations are discussed further in Section 9.

6 Mathematical Modelling

Mathematical modelling is used to describe how an infectious disease spreads through a population over time. In this project, the SIR model is used because it is one of the simplest and most widely used epidemic models.

The SIR model divides the total population into three groups:

- $S(t)$: the number of susceptible individuals who can become infected
- $I(t)$: the number of infected individuals who can spread the disease
- $R(t)$: the number of recovered individuals who are no longer infectious

The total population is assumed to remain constant:

$$N = S(t) + I(t) + R(t)$$

The spread of the disease is described by the following system of differential equations:

$$\begin{aligned}\frac{dS}{dt} &= -\beta SI \\ \frac{dI}{dt} &= \beta SI - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

Here:

- β is the transmission rate of the disease
- γ is the recovery rate
- βSI represents the number of new infections
- γI represents the number of recoveries

The model assumes that susceptible individuals become infected after contact with infected individuals, and infected individuals recover after some period of time.

An important quantity in the SIR model is the basic reproduction number:

$$R_0 = \frac{\beta}{\gamma}$$

The value of R_0 indicates how contagious the disease is:

- If $R_0 > 1$, the disease spreads in the population
- If $R_0 < 1$, the disease gradually disappears
- If $R_0 = 1$, the disease remains stable

One thing worth noting about the SIR equations is that, despite how clean they look, they have no closed-form solution in general. The trouble is the βSI term — it's a product of two unknowns, which makes the system nonlinear, and nonlinear ODEs like this typically can't be solved in terms of familiar functions like exponentials or polynomials.

To see why, it helps to divide the $\dot{I}$ equation by $\dot{S}$, which eliminates time and gives a separable ODE in I and S alone:

$$\frac{dI}{dS} = -1 + \frac{\gamma}{\beta S}$$

Integrating this gives an exact relationship between I and S in the phase plane:

$$I(S) = N - S + \frac{\gamma}{\beta} \ln\left(\frac{S}{S_0}\right)$$

This tells us the *shape* of the epidemic trajectory — for instance, it can be used to find the final epidemic size — but it doesn't tell us where on that curve the system is at any given time t . To get $S(t)$ explicitly, we'd need to solve:

$$\frac{dS}{dt} = -\beta S \left[N - S + \frac{\gamma}{\beta} \ln\left(\frac{S}{S_0}\right) \right]$$

and this ODE, with the logarithm inside, is not analytically integrable. Even Kermack and McKendrick themselves couldn't write down a closed-form $S(t)$ in their original 1927 paper. So numerical methods are not just a convenience here — they're a necessity.

Numerical solution. The approach used in this project is the fourth-order Runge–Kutta (RK4) method. The idea is to step forward through time in increments of size h , approximating the derivative at each step using a weighted average of four slope estimates:

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

where $\mathbf{y} = (S, I, R)^\top$ and the slopes $k_1, \dots, k_4$ are:

$$\begin{aligned} k_1 &= f(t_n, \mathbf{y}_n) \\ k_2 &= f\left(t_n + \frac{h}{2}, \mathbf{y}_n + \frac{h}{2}k_1\right) \\ k_3 &= f\left(t_n + \frac{h}{2}, \mathbf{y}_n + \frac{h}{2}k_2\right) \\ k_4 &= f(t_n + h, \mathbf{y}_n + hk_3) \end{aligned}$$

The method has a global error of $\mathcal{O}(h^4)$, which is accurate enough for epidemic trajectories. A step size of $h = 1$ day was used throughout.

Parameter estimation. Once we can simulate the model numerically, the next step is to find the values of β and γ that make the simulated trajectory match the observed data as closely as possible. If we assume that the observed case counts y_i at time t_i differ from the model predictions $\hat{y}_i$ by independent Gaussian noise with variance σ^2 , the likelihood of seeing the data given parameters (β, γ) is:

$$\mathcal{L}(\beta, \gamma) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \hat{y}_i)^2}{2\sigma^2}\right)$$

Taking the log and dropping constants, maximising the likelihood is equivalent to minimising the sum of squared residuals:

$$\text{SSR}(\beta, \gamma) = \sum_{i=1}^n (y_i - \hat{y}_i(\beta, \gamma))^2$$

Since $\hat{y}_i$ is itself the output of a numerical integrator, there's no closed-form solution for the optimal β and γ either. The minimisation is done iteratively using `scipy.optimize.minimize` with the L-BFGS-B algorithm, which re-runs the ODE solver at each step. The optimisation converged to $\hat{\beta} = 5.0000$ and $\hat{\gamma} = 4.9685$, giving $R_0 \approx 1.01$.

Epidemic control. To model the effect of interventions, a control variable $u(t) \in [0, 1]$ is introduced into the transmission term, representing the fractional reduction in contact rate due to measures like social distancing, masking, or vaccination. The modified system becomes:

$$\frac{dI}{dt} = \beta(1 - u(t)) \frac{SI}{N} - \gamma I$$

The effective reproduction number under this control is:

$$R_{\text{eff}}(t) = \frac{\beta(1 - u(t))}{\gamma} \cdot \frac{S(t)}{N}$$

The epidemic shrinks when $R_{\text{eff}} < 1$, which can be achieved by pushing $u(t)$ high enough (interventions) or by depleting $S(t)/N$ through vaccination.

7 Methodology

The analysis pipeline followed six sequential steps: library setup and data ingestion, preprocessing, exploratory data analysis, SIR model development, model evaluation, and visualization. All code was written in Python and executed in a Jupyter notebook. The key libraries used were NumPy and Pandas for data manipulation, SciPy for numerical integration and optimization, scikit-learn for error metrics, and Matplotlib/Seaborn for visualization.

7.1 Environment Setup

The following libraries were imported at the start of the notebook:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.dates as mdates
import seaborn as sns
from scipy.integrate import odeint
from scipy.optimize import minimize
from sklearn.metrics import mean_absolute_error, mean_squared_error
```

`odeint` is SciPy's wrapper for LSODA, a solver that automatically switches between Adams (non-stiff) and BDF (stiff) methods. It was used to numerically integrate the SIR system. `scipy.optimize.minimize` with the L-BFGS-B method was used for parameter estimation.

7.2 Data Collection and Ingestion

The data was sourced from the Johns Hopkins University COVID-19 repository in its "CONVENIENT" format, where each column represents a country/region and each row a date, with values representing daily new counts (not cumulative). Two separate files were loaded for confirmed cases and deaths:

```
cases_raw = pd.read_csv('CONVENIENT_global_confirmed_cases.csv')
deaths_raw = pd.read_csv('CONVENIENT_global_deaths.csv')
```

7.3 Data Preprocessing

The raw data required substantial cleaning before analysis. The `preprocess()` function below handled the full pipeline in a single pass:

```
def preprocess(df):
    # Row 0 is metadata; rows 1+ are date + daily counts
    dates_col = df.iloc[1:, 0].copy()
    data_part = df.iloc[1:, 1:].copy()

    # Strip sub-national suffixes: 'Australia.1' -> 'Australia'
    countries = [c.split('.')[0].strip() for c in df.columns[1:]]
    data_part.columns = countries
    data_part = data_part.apply(pd.to_numeric, errors='coerce')

    # Aggregate sub-national regions into their parent country
    data_part = data_part.T.groupby(level=0).sum().T

    # Parse dates, sort chronologically, forward-fill sparse gaps
    data_part['Date'] = pd.to_datetime(dates_col.values, format='%m/%d/%y')
    data_part = data_part.sort_values('Date').reset_index(drop=True)
    date_saved = data_part['Date']
    data_part = data_part.drop(columns='Date') \
        .apply(pd.to_numeric, errors='coerce')
    data_part = data_part.ffill().dropna(axis=1, how='all')
    data_part['Date'] = date_saved
    return data_part

daily_cases_df = preprocess(cases_raw)
daily_deaths_df = preprocess(deaths_raw)
```

Key steps in the preprocessing pipeline were: (1) removing the metadata header row; (2) stripping sub-national suffixes (e.g. `Australia.1` → `Australia`) and summing sub-national regions back into their parent country using `groupby`; (3) converting all count columns to numeric, coercing unparseable entries to `NaN`; (4) sorting rows by date and forward-filling sparse gaps with `ffill()`; and (5) dropping any column that is entirely `NaN`.

Global aggregates were then derived by summing across all countries and computing cumulative totals and rolling averages:

```
cc = [c for c in daily_cases_df.columns if c != 'Date']

# Clip negatives (reporting corrections appear as negative daily counts)
daily_cases_global = daily_cases_df[cc].sum(axis=1).clip(lower=0)
daily_deaths_global = daily_deaths_df[dc].sum(axis=1).clip(lower=0)

global_cases = daily_cases_global.cumsum()
global_deaths = daily_deaths_global.cumsum()

# 7-day rolling average to smooth reporting noise
roll_cases = daily_cases_global.rolling(7, min_periods=1).mean()
roll_deaths = daily_deaths_global.rolling(7, min_periods=1).mean()
```

Negative daily counts were clipped to zero to correct for retrospective reporting adjustments, which occasionally introduce negative values in the raw data.

7.4 Exploratory Data Analysis

Summary statistics and key headline figures were computed directly from the aggregated data:

```

cfr = global_deaths.iloc[-1] / global_cases.iloc[-1] * 100
print(f'Total cumulative cases: {global_cases.iloc[-1]/1e6:.1f}M')
print(f'Total cumulative deaths: {global_deaths.iloc[-1]/1e6:.2f}M')
print(f'Case Fatality Rate: {cfr:.2f}%')
print(f'Peak single-day cases: {daily_cases_global.max()/1e6:.2f}M')

```

For the correlation analysis, daily new cases for six major countries (US, India, Brazil, France, Germany, United Kingdom) were differenced and a Pearson correlation matrix was computed:

```

major = ['US', 'India', 'Brazil', 'France', 'Germany', 'United Kingdom']
corr_df = cases[major].diff().clip(lower=0).fillna(0).corr()

mask = np.triu(np.ones_like(corr_df, dtype=bool))
sns.heatmap(corr_df, annot=True, fmt='.2f', cmap='coolwarm',
            vmin=-1, vmax=1, mask=mask)

```

The upper triangle was masked to avoid redundancy in the symmetric matrix.

7.5 SIR Model Development

The SIR system of ODEs was implemented as a Python function compatible with `odeint`:

```

def sir_model(y, t, N, beta, gamma):
    S, I, R = y
    dS = -beta * S * I / N
    dI = beta * S * I / N - gamma * I
    dR = gamma * I
    return dS, dI, dR

```

Parameter estimation was performed by minimizing the Root Mean Squared Error (RMSE) between observed cumulative cases and the model's $I(t) + R(t)$ trajectory over the first 200 days. Note that $I + R$ equals the total number of individuals who have ever been infected, which corresponds to the cumulative confirmed case count.

```

def fit_sir(cumulative_infected, N, n_days=200):
    obs = cumulative_infected[:n_days].values
    t = np.arange(n_days)

    I0 = obs[0] if obs[0] > 0 else 1
    y0 = N - I0, I0, 0 # (S0, I0, R0)

    def loss(params):
        beta, gamma = params
        if beta <= 0 or gamma <= 0:
            return 1e12
        sol = odeint(sir_model, y0, t, args=(N, beta, gamma))
        I_pred = sol[:, 1] + sol[:, 2] # cumulative = I + R
        return np.sqrt(np.mean((I_pred - obs) ** 2))

    result = minimize(loss, x0=[0.3, 0.1],
                     bounds=[(1e-4, 10), (1e-4, 10)],
                     method='L-BFGS-B')
    return result.x # returns (beta_fit, gamma_fit)

N_global = 7_900_000_000
beta_fit, gamma_fit = fit_sir(global_cases, N_global, n_days=200)

```

At each iteration of the optimizer, `odeint` numerically integrates the SIR system forward in time and the resulting trajectory is compared against the observed data. The L-BFGS-B method uses gradient approximations (limited-memory Broyden-Fletcher-Goldfarb-Shanno with box constraints) to efficiently navigate

the parameter space. Initial guesses were set to $\beta_0 = 0.3$ and $\gamma_0 = 0.1$, with both parameters bounded to $(10^{-4}, 10)$.

7.6 Model Evaluation

Four standard error metrics were computed to quantify the quality of the SIR fit:

```
from sklearn.metrics import mean_absolute_error, mean_squared_error

mae = mean_absolute_error(obs, cum_pred)
mse = mean_squared_error(obs, cum_pred)
rmse = np.sqrt(mse)
mape = np.mean(np.abs((obs - cum_pred) / (obs + 1))) * 100

print(f'MAE: {mae:.0f}')
print(f'MSE: {mse:.0f}')
print(f'RMSE: {rmse:.0f}')
print(f'MAPE: {mape:.0f}%')
```

MAE and RMSE are both in units of cases (not squared), making them directly interpretable. RMSE penalizes large errors more heavily than MAE due to the squaring. MAPE is scale-free (percentage), but the +1 in the denominator guards against division by zero on the earliest days when observed counts are near zero. Residuals were also plotted over time to diagnose any systematic model misfit (see Figure 7).

8 Results

8.1 Global Cumulative Trends

Figure 1 shows the global cumulative confirmed cases and deaths from December 2019 to the end of 2022. The total confirmed cases reached approximately 670 million, while cumulative deaths approached 6.9 million. The curves exhibit multiple phases of accelerated growth corresponding to successive waves driven by new variants.

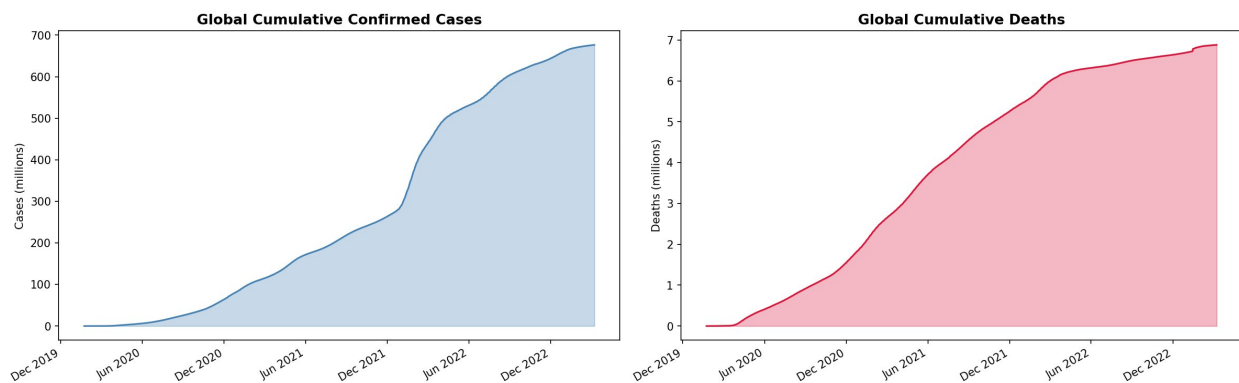


Figure 1: Global cumulative confirmed COVID-19 cases (left) and deaths (right) from December 2019 to early 2023.

8.2 Daily New Cases and Deaths

Figure 2 presents the daily new cases and deaths (7-day moving average) over the same period. Distinct epidemic waves are visible, with the largest peak in daily cases occurring around January 2022 (approximately 3.5 million cases per day), driven by the Omicron variant. Daily deaths peaked earlier, during the Delta wave in late 2021.

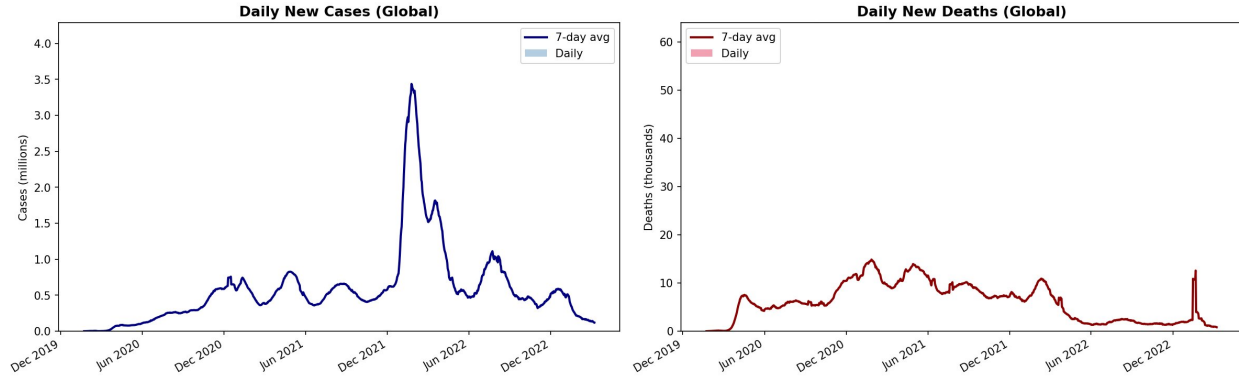


Figure 2: Daily new COVID-19 cases (left) and deaths (right) globally. Solid lines represent 7-day moving averages.

8.3 Country-Level Distribution

Figure 3 ranks the top 10 countries by total confirmed cases. The United States recorded the highest burden with approximately 103.8 million cases, followed by India (44.7M), France (39.9M), and Germany (38.2M). This distribution reflects both the scale of outbreaks and the level of testing infrastructure in each country.

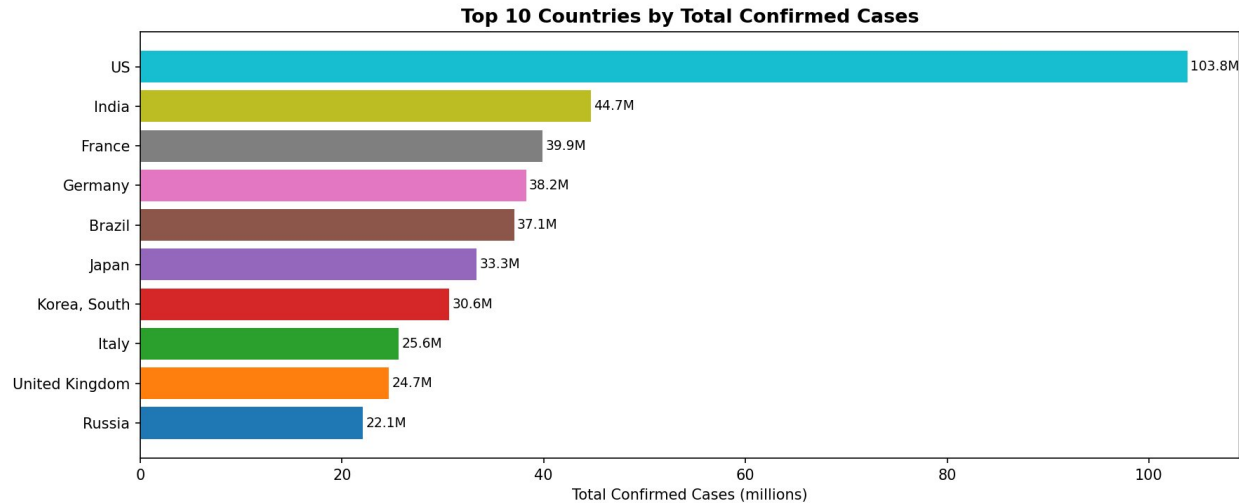


Figure 3: Top 10 countries by total confirmed COVID-19 cases (in millions).

8.4 Cross-Country Correlation

Figure 4 displays a pairwise correlation matrix of daily new cases across six major countries: US, India, Brazil, France, Germany, and United Kingdom. The strongest correlation is observed between the US and France (0.62) and between the US and the United Kingdom (0.57), suggesting similar epidemic timing or shared variant introduction pathways. India and Germany show a near-zero correlation (-0.02), reflecting distinct epidemic dynamics.

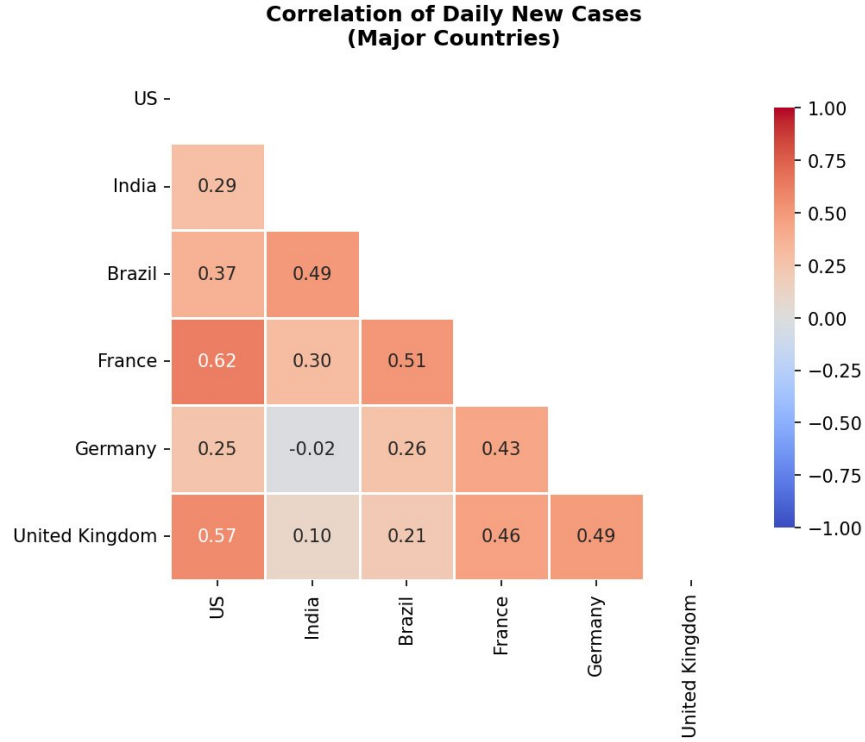


Figure 4: Pairwise Pearson correlation of daily new COVID-19 cases among major countries.

8.5 SIR Model Fit

The SIR model was fitted to the first 300 days of the global epidemic (February–December 2020) using nonlinear least squares optimization. The estimated parameters are:

$$\hat{\beta} = 5.0000, \quad \hat{\gamma} = 4.9685, \quad R_0 = \frac{\hat{\beta}}{\hat{\gamma}} \approx 1.01$$

Figure 5 compares the model prediction (orange curve) with observed cumulative cases (blue dots). The model captures the general exponential growth trend but diverges notably in the mid-period (May–September 2020), where the observed data grows faster than the single-wave SIR trajectory can accommodate.

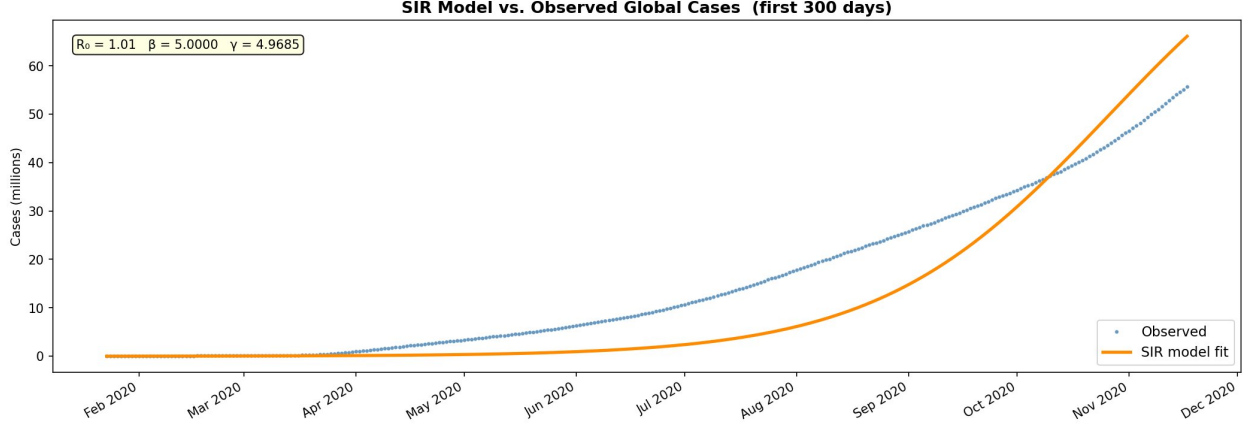


Figure 5: SIR model fit versus observed global cumulative cases for the first 300 days of the pandemic. Estimated $R_0 = 1.01$.

8.6 SIR Compartment Trajectories

Figure 6 shows the time evolution of all three SIR compartments under the fitted parameters. The susceptible population (S) remains approximately constant at around 7.7 billion throughout, as only a small fraction of the global population had been infected by December 2020. The infectious compartment (I) remains small in magnitude, while the removed compartment (R) grows steeply toward the end of the period.

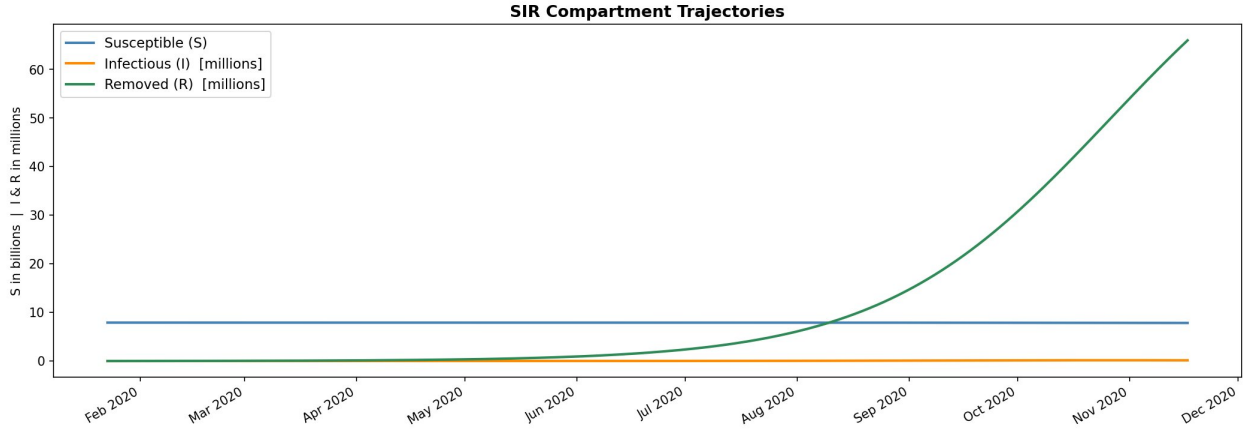


Figure 6: SIR compartment trajectories: Susceptible (S , in billions), Infectious (I , in millions), and Removed (R , in millions) over the first 300 days.

8.7 Residual Analysis

Figure 7 presents the residuals (observed minus predicted) over time and their distribution. The residual plot reveals a clear systematic pattern: the model underestimates cases from approximately April to September 2020 (positive residuals peaking at +12 million), then overestimates from October onward (negative residuals reaching -10 million by December 2020). This non-random structure indicates that the basic SIR model does not fully capture the multi-wave dynamics of the pandemic. The residual distribution is bimodal rather than centered at zero, further confirming model misspecification.

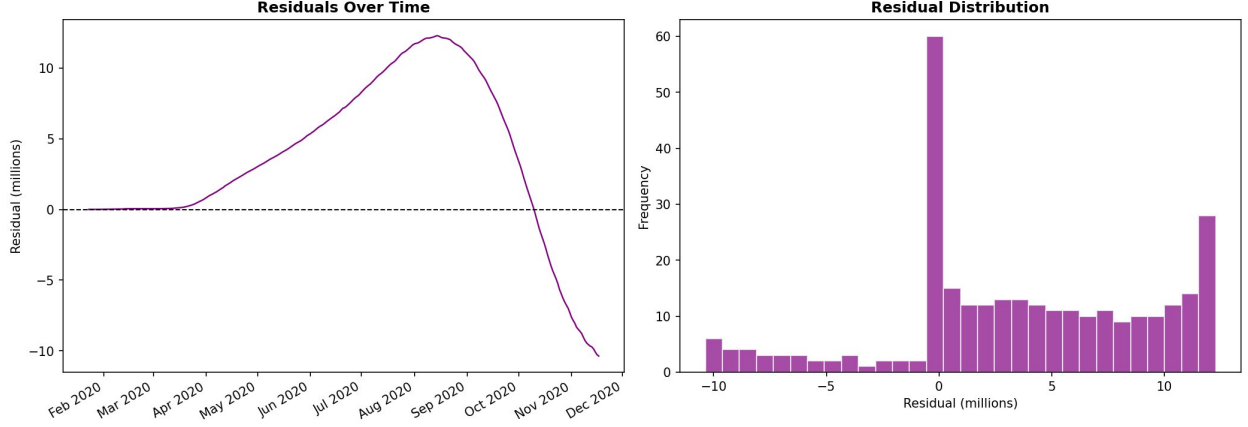


Figure 7: Residuals over time (left) and residual distribution histogram (right) for the SIR model fit.

9 Discussion

Exploratory findings. The exploratory analysis confirms the multi-wave structure of the COVID-19 pandemic. Four distinct surges are visible in the daily case data (Figure 2): an initial wave in early 2020, a second wave in late 2020, a third driven by the Delta variant in mid-2021, and the dominant Omicron surge in January 2022 which dwarfed all prior peaks in case counts but produced comparatively fewer deaths per case, reflecting both the variant’s reduced intrinsic severity and widespread population immunity from prior infection and vaccination.

The dominance of high-income countries (US, France, Germany, UK) in the top-10 case count ranking (Figure 3) partly reflects superior testing infrastructure rather than a higher true infection burden. Low- and middle-income countries almost certainly experienced higher absolute infection numbers but had far less diagnostic capacity to detect and report them. This systematic undercount must be borne in mind when interpreting all case-based analyses.

The cross-country Pearson correlations (Figure 4) are moderate (0.21–0.62) rather than high, which is epidemiologically informative. It suggests that while international travel seeded variants globally — producing broad synchrony across waves — the precise timing and height of each surge within a country was shaped primarily by local factors: vaccine uptake, non-pharmaceutical interventions, prior immunity, and healthcare capacity.

Model performance and limitations. The fitted parameters $\hat{\beta} = 5.0000$ and $\hat{\gamma} = 4.9685$ yield $R_0 \approx 1.01$. This value is far below published estimates for COVID-19 ($R_0 \approx 2.2$ –3.6 for the original strain [5]), and the discrepancy is expected. The basic SIR model assumes a homogeneous, well-mixed population with constant transmission and recovery rates. In reality, the global aggregate data reflects heterogeneous populations across 201 countries, time-varying government interventions, immunity accumulation, successive variant replacements, and large cross-country differences in testing. The fitted β and γ are therefore effective averages across all these sources of variation, not true biological parameters.

The residual analysis (Figure 7) makes the misfit explicit: residuals follow a systematic arc rather than random scatter. The model underestimates cases from April to September 2020 as the pandemic spread rapidly to new regions, then overestimates from October 2020 onward as non-pharmaceutical interventions and eventual vaccination slowed transmission below what the single-wave trajectory predicted. A bimodal residual distribution rather than a unimodal near-Gaussian one further confirms model misspecification.

These limitations point directly toward natural extensions: (1) a *time-varying* $\beta(t)$ could absorb the effect of changing interventions; (2) the SEIR model with an exposed compartment better reflects COVID-19’s incubation period; (3) an SEIRV model with a vaccinated compartment would capture the immunity dynamics of 2021–2022; and (4) fitting country-specific rather than global data would remove the averaging artefact.

Intervention analysis. Despite the fitting limitations, the control framework is theoretically sound. With the modified infection term $\beta(1 - u(t))SI/N$, the effective reproduction number becomes:

$$R_{\text{eff}}(t) = \frac{\beta(1 - u(t))}{\gamma} \cdot \frac{S(t)}{N}$$

Epidemic suppression requires $R_{\text{eff}}(t) < 1$, which is achievable either by reducing β through non-pharmaceutical interventions (masks, distancing, ventilation) or by reducing $S(t)/N$ through vaccination. For $R_0 \approx 1.01$, even a very small control effort $u > 0.01$ suffices to suppress the epidemic — but for the published COVID-19 estimate of $R_0 \approx 3$, a control effort $u > 0.67$ (i.e., a 67% reduction in transmission) would be required, consistent with empirical findings on the effectiveness of strict lockdowns [11].

10 Conclusion

This project applied the SIR epidemic model to global COVID-19 data from the JHU CSSE repository to estimate transmission parameters, characterise epidemic dynamics through exploratory analysis, and study the theoretical impact of control measures. The principal findings are:

- The global pandemic exhibited a clear multi-wave structure, with cumulative confirmed cases reaching ~ 670 million and deaths ~ 6.9 million by March 2023. The largest single-day case peak (~ 3.5 million) occurred during the Omicron wave in January 2022.
- The United States reported the highest total confirmed cases (103.8 million), though case counts in high-income countries reflect testing capacity as much as true infection burden.
- Cross-country Pearson correlations in daily case counts ranged from -0.02 (India–Germany) to 0.62 (US–France), indicating partial global synchrony with substantial local heterogeneity.
- Nonlinear least squares fitting of the SIR model yielded $\hat{\beta} = 5.0000$, $\hat{\gamma} = 4.9685$, and $R_0 \approx 1.01$. Residual analysis revealed systematic non-random misfit, attributable to the model’s inability to represent multiple waves, time-varying interventions, and evolving population immunity within a single homogeneous framework.
- Introduction of a control variable $u(t)$ demonstrates that reducing the transmission rate by a factor $(1 - u)$ lowers the effective reproduction number proportionally. For the global aggregate fit, epidemic suppression requires only $u > 0.01$; for the more realistic country-level $R_0 \approx 3$, a reduction of at least 67% in transmission is required.

Future directions. Several natural extensions would strengthen this analysis. Replacing the basic SIR with an SEIR or SEIRV (vaccinated) model would improve biological realism. Introducing a time-varying transmission rate $\beta(t)$ — fit piecewise or as a smooth spline — would allow the model to track intervention changes. Applying the framework to individual countries rather than the global aggregate would remove the cross-country averaging artefact and yield epidemiologically interpretable parameter estimates. Finally, a formal optimal control analysis using Pontryagin’s maximum principle would provide rigorous, quantitative guidance on the most cost-effective intervention schedule.

References

- [1] W. O. Kermack and A. G. McKendrick, *A contribution to the mathematical theory of epidemics*, Proceedings of the Royal Society of London A, 115(772):700–721, 1927.
- [2] E. Dong, H. Du, and L. Gardner, *An interactive web-based dashboard to track COVID-19 in real time*, The Lancet Infectious Diseases, 20(5):533–534, 2020.
- [3] S. A. Lauer et al., *The incubation period of coronavirus disease 2019 (COVID-19) from publicly reported confirmed cases*, Annals of Internal Medicine, 172(9):577–582, 2020.

- [4] R. M. Anderson and R. M. May, *Infectious Diseases of Humans: Dynamics and Control*, Oxford University Press, 1992.
- [5] Y. Liu, A. A. Gayle, A. Wilder-Smith, and J. Rocklöv, *The reproductive number of COVID-19 is higher compared to SARS coronavirus*, Journal of Travel Medicine, 27(2), 2020.
- [6] Y. Liu and J. Rocklöv, *The effective reproductive number of the Omicron variant of SARS-CoV-2 is several times relative to Delta*, Journal of Travel Medicine, 29(3), 2022.
- [7] J. R. Dormand and P. J. Prince, *A family of embedded Runge-Kutta formulae*, Journal of Computational and Applied Mathematics, 6(1):19–26, 1980.
- [8] A. C. Hindmarsh, *ODEPACK, a systemized collection of ODE solvers*, IMACS Transactions on Scientific Computing, 1:55–64, 1983.
- [9] S. P. Sethi and P. W. Staats, *Optimal control of some simple deterministic epidemic models*, Journal of the Operational Research Society, 29(2):129–136, 1978.
- [10] D. H. Morris, F. W. Rossine, J. B. Plotkin, and S. A. Levin, *Optimal, near-optimal, and robust epidemic control*, Communications Physics, 4:78, 2021.
- [11] S. Flaxman et al., *Estimating the effects of non-pharmaceutical interventions on COVID-19 in Europe*, Nature, 584:257–261, 2020.